

MMAHULO BERNARD MOLABA

COMPUTER SYSTEMS ENGINEER & FULL-STACK IOT DEVELOPER

Phone: 071 503 2734 Email: mmahulo.molaba@gmail.com Location: Midrand / Juksei View, Gauteng
License: Code C10

PROFESSIONAL SUMMARY

Final-year **Computer Systems Engineering** student at Tshwane University of Technology seeking a Work-Integrated Learning (WIL) placement or Graduate Engineer role. Specialized in bridging **embedded hardware architecture** (AVR, ESP32, Proteus, PLC, I2C/UART protocols) with **modern full-stack cloud software** (Next.js, React, TypeScript, Supabase, Tailwind CSS). Demonstrates strong hands-on capability in designing end-to-end IoT access control systems, autonomous robotics (ROS2), and industrial automation.

EDUCATION

National Diploma: Computer Systems Engineering (In Progress) Expected Completion: 2026
Tshwane University of Technology (TUT) — Pretoria, South Africa

TECHNICAL EXPERTISE

Web & Cloud Stack	Next.js (React), TypeScript, Node.js, Supabase (PostgreSQL / REST APIs), Tailwind CSS, shadcn/ui
Embedded Systems & IoT	C / C++, Microcontroller Architecture (ATmega328P, ESP32 / ESP32-CAM), I2C, UART, SPI, Serial Communication
Simulation & EDA Tools	Proteus Design Suite (Circuit Simulation & Layout), Arduino IDE, AVR-GCC Toolchain
Automation & Robotics	PLC Programming (Siemens TIA Portal S7-1200, Allen Bradley), ROS2 Jellyfish (Autonomous Navigation), Sensors
Programming Languages	C++, C, Python, TypeScript, JavaScript, SQL, Assembly
Testing & Tools	Jest, PyTest, GoogleTest, Git, GitHub, Linux CLI, SSH

FEATURED ENGINEERING PROJECTS

Access Control & Asset Management System (ACAMS) 2026
Tech Stack: Dual ATmega328P, ESP32-CAM, I2C/UART Protocols, Next.js, Supabase, Proteus, Tailwind CSS

- Engineered an end-to-end multi-factor physical access control and electronic hardware asset tracker for laboratory facilities.
- Designed and simulated dual ATmega328P microcontroller communication using **I2C** for local peripheral sync and **UART** for serial payload transmission to the ESP32 Wi-Fi module.
- Built a full-stack Next.js and Supabase administrative dashboard providing real-time access logs, user privilege control, and equipment checkout tracking.
- Pre-validated circuit schematics, relay driver actuation, and sensor responses using **Proteus Design Suite** prior to hardware assembly.

ROS2 Autonomous Wall Avoidance Mobile Robot 2025
Tech Stack: ROS2 Jellyfish, Python, C++, Ultrasonic Sensors, Linux

- Developed autonomous robotic navigation nodes utilizing ROS2 middleware for obstacle detection and real-time path adjustments.
- Integrated range sensor streams and calibrated motor control drive signals for smooth collision avoidance algorithms.

Industrial PLC Conveyor Sorting & Automation System Tech Stack: Siemens TIA Portal (S7 PLC), Ladder Logic, Proximity Sensors	2025
• Programmed Ladder Logic routines for automated material separation and conveyor sorting based on sensor feedback. • Implemented fault-detection, emergency stop logic, and cycle tracking to ensure industrial safety compliance.	

Unity Machine Learning Reinforcement Agent Tech Stack: Unity ML-Agents, C#, Python, Reinforcement Learning	2025
• Trained an autonomous AI agent using Proximal Policy Optimization (PPO) for complex object interaction within virtual environments.	

ADDITIONAL COMPETENCIES & CREDENTIALS

- **Driver's License:** Code C10 (Clean record)
- **Soft Skills:** Hardware-Software Troubleshooting, Systems Integration, Technical Documentation, Agile Team Collaboration
- **Languages:** English (Fluent), Native South African Languages

REFERENCES

Academic and professional references available immediately upon request.